Biochemistry Cumulative Exam
February 17, 2012

(Supplemental Information Attached)

A mouse is fed a diet including the amino acid glutamate, but no carbohydrate, or fatty acids. Small amounts of the essential amino acids and the necessary vitamins and minerals are also included in the diet.

1. (50) Can the mouse synthesize the amino acid alanine, NH_2-$CH(CH_3)$-COOH, on this diet using glutamate as the sole source of carbon, nitrogen and energy? If the answer is yes, then outline a metabolic pathway for this process. How many glutamate molecules must be used to provide enough carbon atoms, nitrogen, ATP, and NADH molecules to synthesize each molecule of alanine? Write a complete, balanced equation for this process giving the numbers of ATP, NADH, and CO_2 generated.

2. (50) Can the mouse synthesize glucose on this diet using glutamate as the sole source of carbon and energy? If the answer is yes, then outline a metabolic pathway for this process. How many glutamate molecules must be used to provide enough carbon atoms, ATP, and NADH molecules to synthesize each molecule of glucose? (Assume that NADH can freely move between the mitochondria and the cytoplasm.) Write a complete, balanced equation for this process giving the numbers of ATP, NADH, and CO_2 generated.

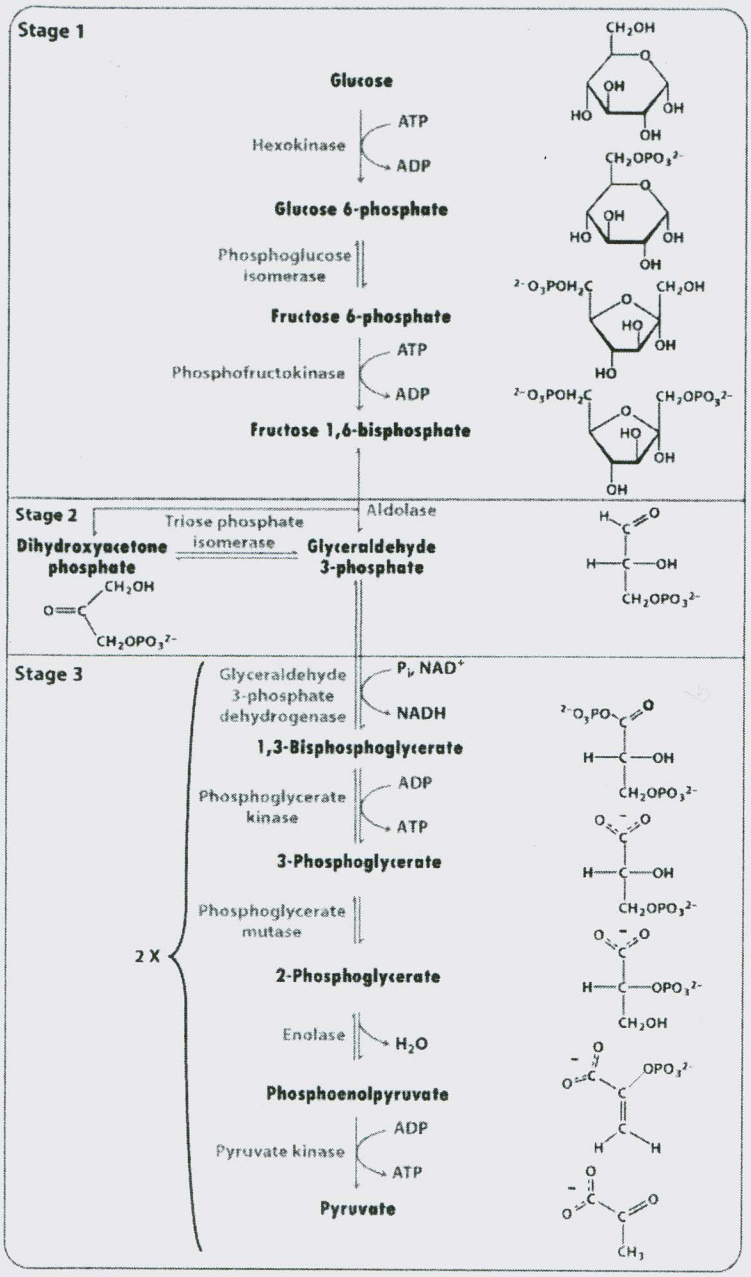

Figure 16-2
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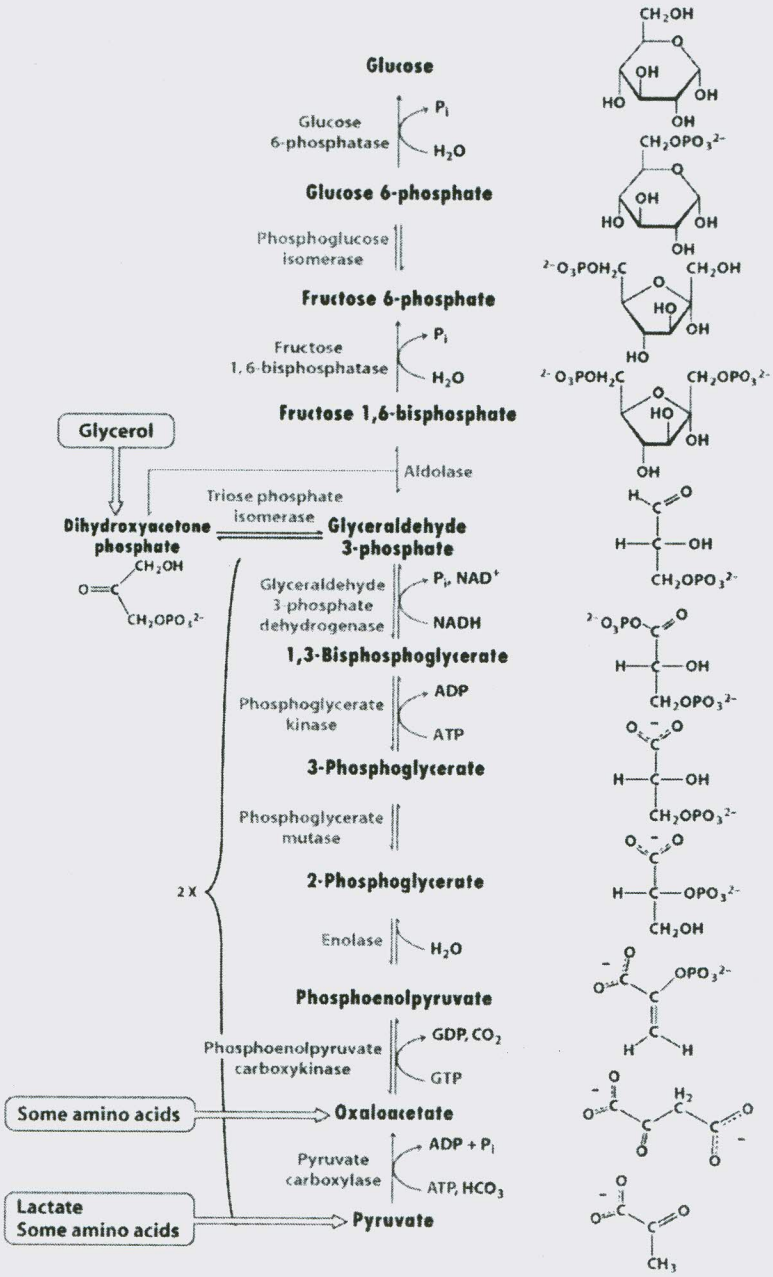

Figure 16-22
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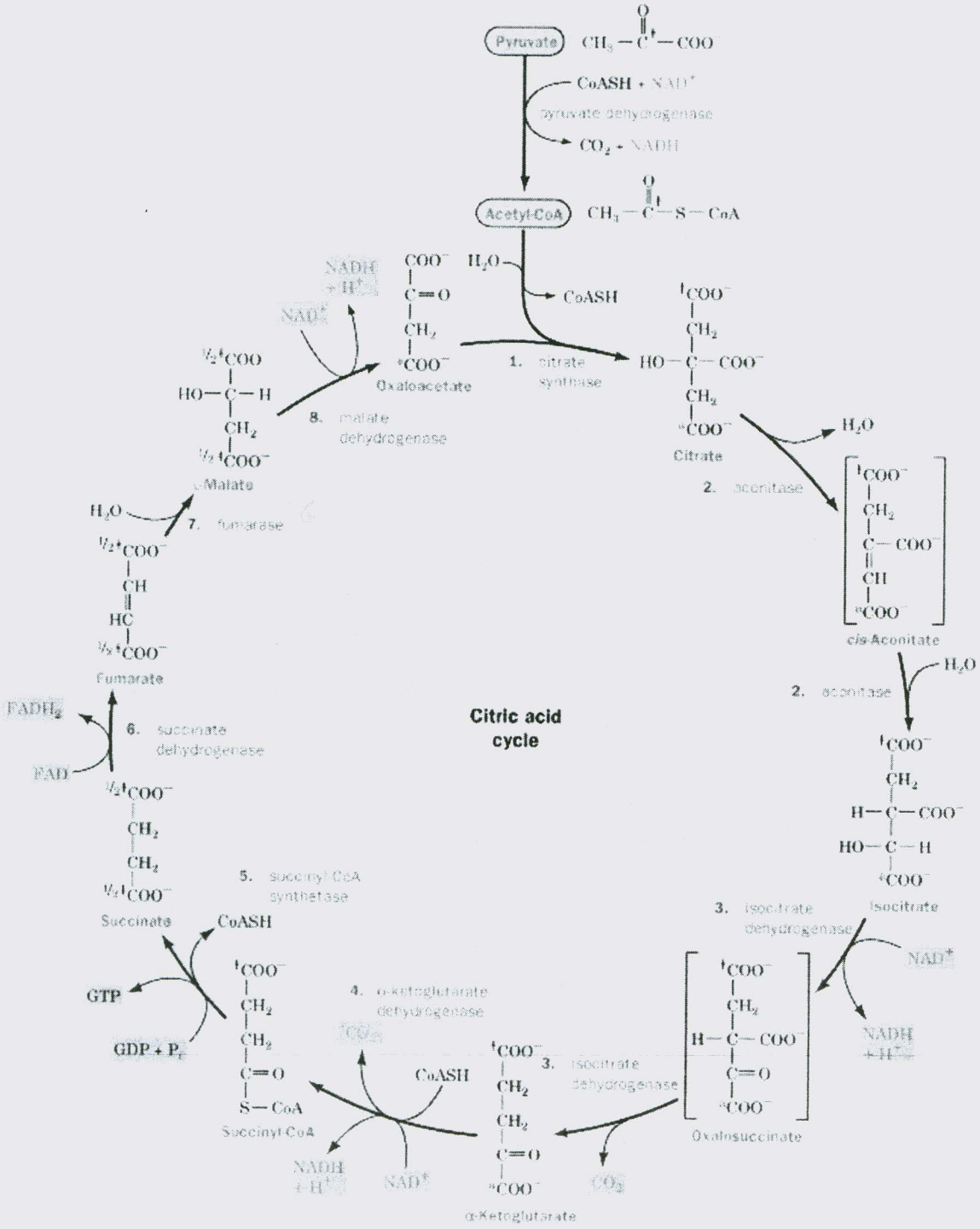
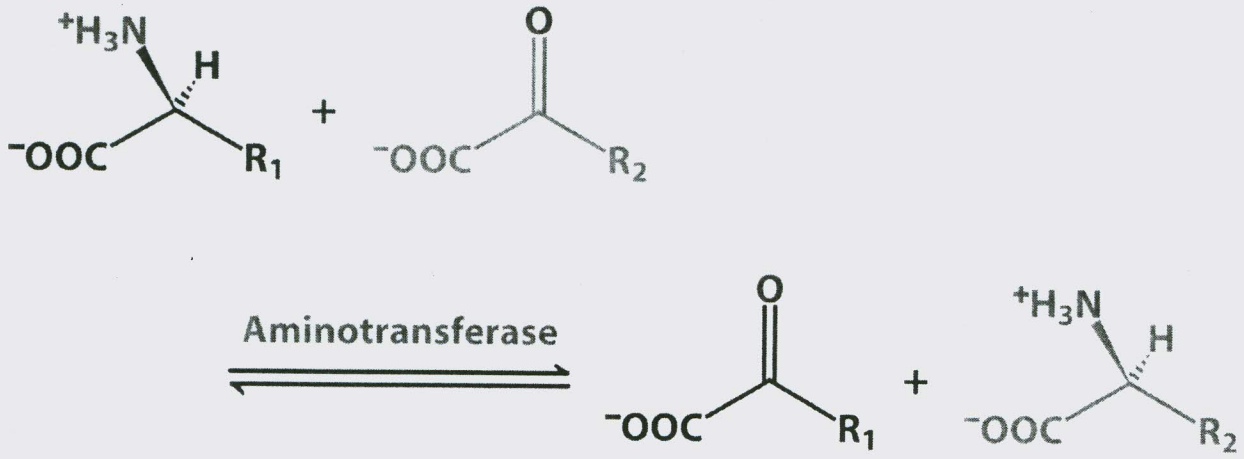

Citric acid cycle
Pyruvate
CoASH + NAD+
pyruvate dehydrogenase
CO2 + NADH
Acetyl-CoA
H2O
CoASH
1. citrate synthase
Oxaloacetate
Citrate
2. aconitase
H2O
cis-Aconitate
2. aconitase
H2O
Isocitrate
3. isocitrate dehydrogenase
NAD+
NADH + H+
Oxalosuccinate
3. isocitrate dehydrogenase
CO2
α-Ketoglutarate
4. α-ketoglutarate dehydrogenase
CO2
CoASH
NADH + H+
NAD+
Succinyl-CoA
5. succinyl-CoA synthetase
CoASH
GTP
GDP + Pi
Succinate
6. succinate dehydrogenase
FADH2
FAD
Fumarate
7. fumarase
H2O
Malate
8. malate dehydrogenase
NADH + H+
NAD+



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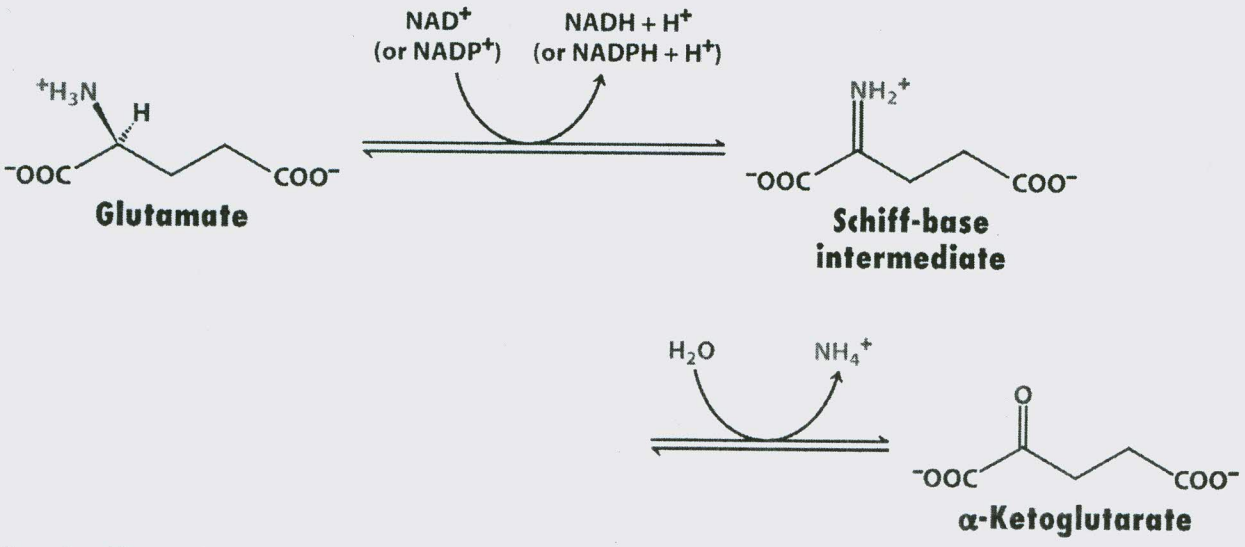

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